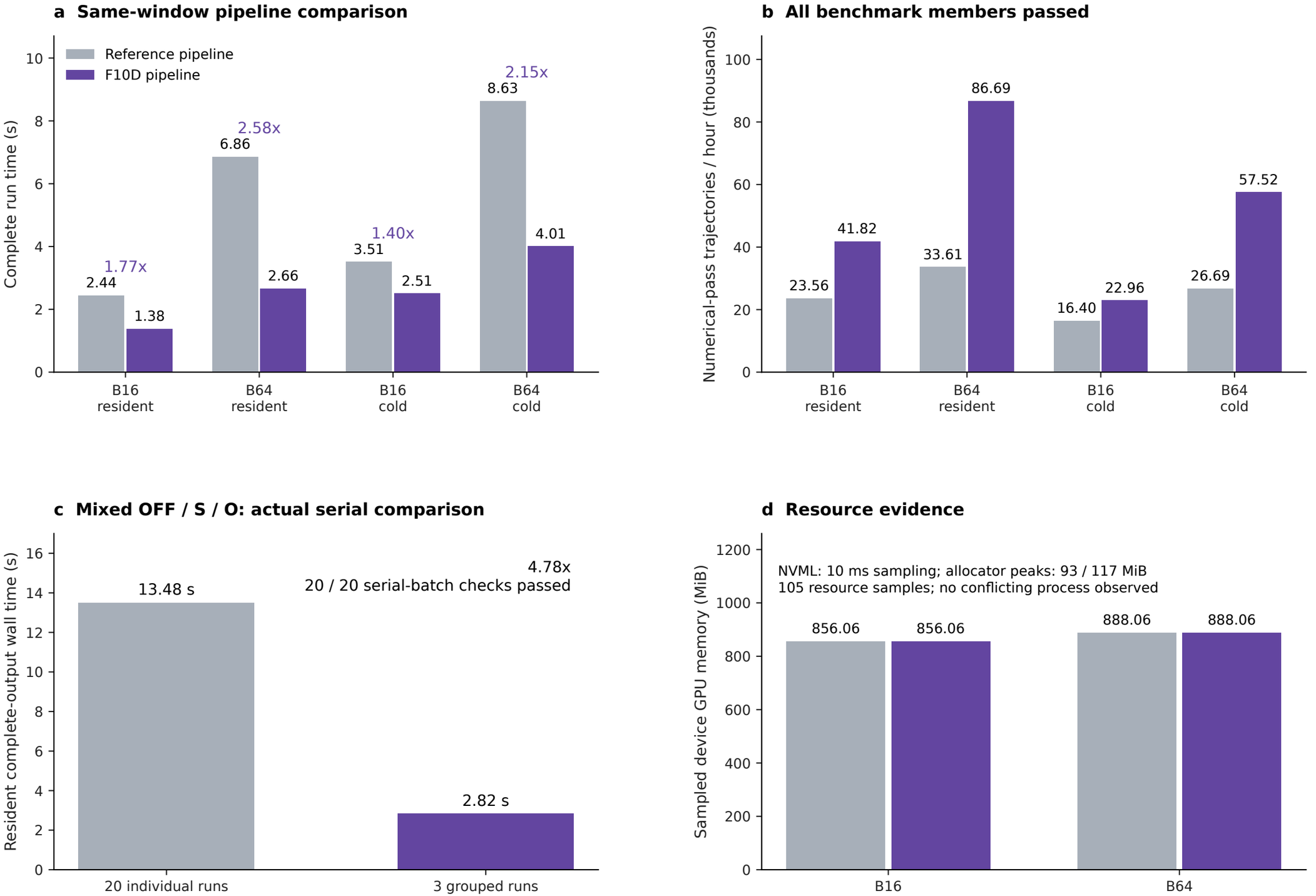

All runs: full-particle M0, 10 μs steps, 201 frames, unchanged quality budgets. B64 cold speedup: 2.15x (target >= 1.25x).
One matched pair per size/startup; BENCHMARK_ONLY. Hourly rates are short-run extrapolations, not independent scientific samples.